

BADRE ALAM

Electronics Engineer | VLSI | RTL Design | Verilog | FPGA

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SKILLS

- Verilog HDL
- VLSI Design
- FPGA Development
- RTL Design
- FSM Design
- Testbench Verification
- SystemVerilog
- Xilinx Vivado 2022.2
- EDA Tools
- Python
- C / C++
- Tcl Scripting
- MATLAB
- Git
- Linux CLI
- VS Code

STRENGTHS

Strong foundation in RTL Design, Verilog HDL, FPGA Development, FSM Design, and Testbench Verification — directly applicable to Display IC and semiconductor engineering.

EDUCATION

B.Tech — ECE

Rajkiya Engineering College
Kannauj | Sep 2023 – Sep 2027
CGPA: 7.46 / 10.0

Intermediate (Class XII)

UP Board, Sant Kabir Nagar
2022 | 74.8%

High School (Class X)

UP Board, Sant Kabir Nagar
2020 | 79.33%

CERTIFICATIONS

- VLSI Design**
Internshala (2025)
- Gold Developer Certificate**
IIT Kanpur (2025)

CONTACT

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Sant Kabir Nagar, U.P.

SUMMARY

Electronics & Communication Engineering student (CGPA 7.46) with hands-on expertise in RTL design, Verilog HDL, FSM-based digital control systems, and FPGA implementation on Xilinx Vivado. Completed 4 independent hardware projects covering precision arithmetic units, sequential controllers, and signal processing modules — achieving design accuracy within ± 0.02 error margins. Strong foundation in display IC principles including gamma correction logic, pixel-level signal control, Timing Controller (T-CON) architecture, and OLED/AMOLED panel driver design. VLSI Design Certified; Gold Developer Certificate awarded by IIT Kanpur.

EXPERIENCE

VLSI Design Intern

Internshala

Aug 2025 – Oct 2025

2-week Online VLSI Training Program | Remote

- Completed structured 2-week VLSI design training covering RTL design methodology, ASIC/FPGA design flow, and semiconductor IC fundamentals with industry-standard tools.
- Earned the Internshala VLSI Design Certificate upon successfully completing all training modules and final evaluation.

Gold Developer

IIT Kanpur

May 2025 – Jun 2025

1-week Virtual Lab Bootcamp | Kanpur, U.P.

- Completed a 1-week intensive Virtual Lab bootcamp covering advanced electronics experiments and digital system simulation.
- Awarded the Gold Developer Certificate — highest performance tier — for exceptional scores across all lab modules and assessments.

PROJECTS

Activation Function Unit (AFU) — Sigmoid Hardware Block

Aug 2025

Verilog HDL | Xilinx Vivado 2022.2 | Digital Arithmetic | FPGA

- Designed a hardware sigmoid AFU in Verilog implementing nonlinear approximation for pixel brightness calibration and gamma correction in display pipeline ICs — achieving ± 0.02 maximum error across 50+ test vectors.
- Optimized RTL for minimal LUT utilization through piecewise approximation — essential in area-constrained display driver ICs for OLED/AMOLED panels.

Traffic Light Controller — FSM-Based Timing Controller

Apr 2025

Verilog HDL | FSM Design | Testbench Verification | Xilinx Vivado

- Implemented a 4-state FSM Traffic Light Controller in Verilog with configurable timing delays — the same architecture used in T-CON chips sequencing gate and source driver signals in display panels.
- Authored a testbench with 30+ directed test cases achieving 100% state coverage; identified 2 timing edge-case bugs before synthesis.

Sequence Detector (1101) — Serial Pattern Recognition

Feb 2025

Verilog HDL | Moore FSM | Xilinx Vivado

- Built a Moore FSM-based serial sequence detector for binary pattern 1101 — applicable to display data protocol parsing (MIPI DSI sync detection).
- Verified across all 16 possible 4-bit input patterns with zero false positives and zero missed detections.

Automatic Washing Machine Controller — Multi-Stage FSM

Oct 2024

Verilog HDL | Sequential FSM | Xilinx Vivado

- Designed a 5-state FSM (Idle, Fill, Wash, Drain, Spin) with parameterized timed transitions — analogous to display driver IC power-on sequencing, verified across 3 operational profiles with a reusable testbench.